

DIVYANSHU TANDON

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EDUCATION

Arizona State University (ASU), Tempe, Arizona

Expected May 2027

Bachelor of Science in Computer Science

Data Structures and Algorithms, Operating Systems, Object-Oriented Programming, Database Management, Software QA and Testing

SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C, C++

Web & Backend: React, FastAPI, REST API Design, Vite, Tailwind CSS, HTML/CSS, SQLite

Scientific Computing: NumPy, pandas, h5py, PyQt/PySide, Image Processing, Cross-Correlation Analysis

Platforms & Tools: Git/GitHub/GitLab, Linux, Bash, GitHub Actions (CI/CD), Reddit Devvit, Redis, Docker, dbt, Airflow

WORK EXPERIENCE

Research Assistant, BDI Biodesign - Center for Applied Structural Discovery, ASU

Sept 2025 - Present

- Rebuilt a single-file Python script into an automated pipeline that processes entire folders of experiment data into Excel reports.
- Added a validation step comparing two calculation methods against manual measurements to confirm accuracy

Undergraduate Researcher, CXFEL (Compact X-ray Free Electron Laser), ASU

May 2024 - Present

- Built Python tools(NumPy & PyQt) to process and visualize X-ray image data for studying the structure of biological molecules.
- Wrote image-processing algorithms and automated pipelines that sped up analysis and calibrated imaging data.

Summer Researcher, SLAC National Accelerator Laboratory, Menlo Park, California

July 2025

- Operated and fine-tuned equipment that delivers liquid samples into an X-ray beam at LCLS(Linac Coherent Light Source), keeping delivery stable and measurements clean for high-concentration biological samples.
- Worked with a team from multiple universities and labs to collect data in real time, successfully measuring particles as small as 30 nanometers, roughly the size of a small virus.

Data Analyst, MaRTy Fieldwork and Urban Data Collection, ASU

May 2024 - July 2024

- Led urban fieldwork data collection using mobile MaRTy sensor platforms for shade and heat-exposure analysis.
- Applied statistical analysis to validate temperature data and synthesized findings for ASU's Shade Lab urban heat planning.

PROJECTS

Reborn - Scientific Visualization Framework (Open-Source, Gitlab)

July 2024 - Present

- Redesigned the detector Geometry Editor around a delegate/strategy pattern (abstract base class, concrete PyQt implementation) to decouple rotation/translation math from the UI, and fixed a GitLab-tracked crash from zero/negative detector distances with input validation across both code paths.
- Built a custom validated spin-box widget and resolution/q-value ring labeling; added an undo/redo history stack and threshold-based masking to the pixel-mask editor.

The Guardian - MCP Security Gateway (Python, MCP, Docker)

Summer 2026

- Engineered a security gateway for AI agents that automatically blocks unauthorized tool access and detects hidden malicious instructions in real time, using detection rules based on well-known AI security risk categories.
- Deployed the system as two independent Docker containers with a live dashboard for monitoring activity in real time, backed by a GitHub Actions pipeline that automatically runs 72 tests before every deployment.

DataDoc - Metadata-Aware Code Generation Agent (Python, Claude Agent SDK, DataHub MCP)

Summer 2026

- Built an AI agent on Anthropic's Claude Agent SDK that queries a live DataHub catalog over the Model Context Protocol for real schema and lineage before generating dbt models, Airflow DAGs, and SQL migrations, eliminating hallucinated column names.
- Implemented closed-loop metadata write-back (agent tags undocumented/PII columns back to DataHub mid-generation) and validated It ends-to-end against a self-hosted Dockerized DataHub instance by independently querying the catalog's GraphQL API to confirm Each write actually persisted.

ReddWall - Reddit Devvit Moderation App (TypeScript, AI-assisted development, Redis)

Summer 2026

- Built and published a Devvit app giving moderators native tools to preview, lock, and remove risky comment threads during raids, brigading, and spam bursts.
- Implemented phrase-based targeting, protected-comment safeguards, and Redis-backed moderation history, with a full lint, type-check, test, and build pipeline.

PUBLICATIONS

- Ansari, A. et al. (incl. **Divyanshu Tandon**) “Synchronized droplet nozzle for in-vacuum X-ray scattering experiments,” Lab on a Chip, 2025, 25, 6465, DOI: 10.1039/D5LC00063G.
- Two first-author research posters on GDVN nozzle design and characterization for XFEL sample delivery, ASU Biodesign Institute.